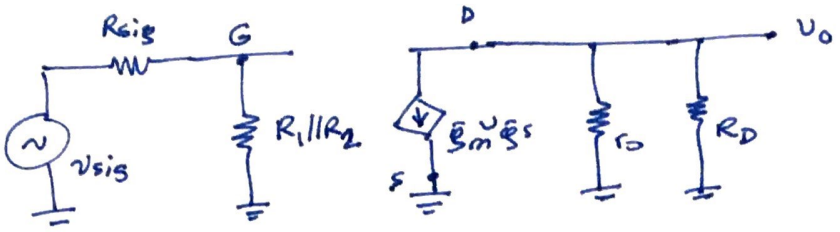
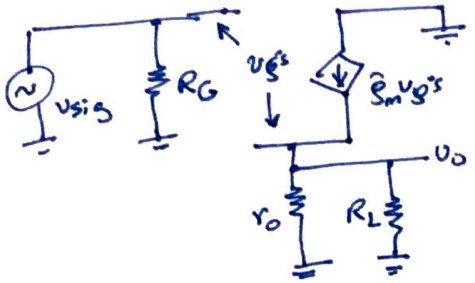
Electronic I Quiz2(2) solution

1)



2)



$v_i = v_{gs} + v_o$

$v_o = g_m v_{gs} R_L$

ignore r_o

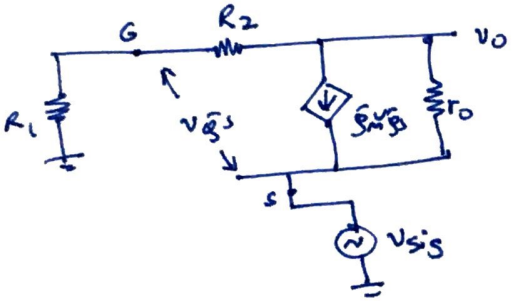
$$A_v = \frac{g_m R_L}{1 + g_m R_L}$$

$R_{in} = R_G$

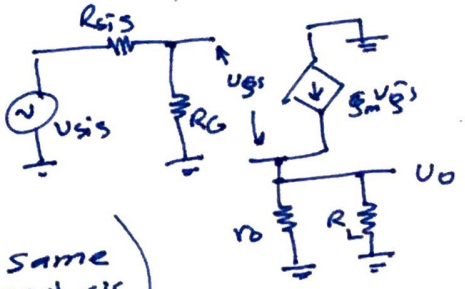
$R_o = 1/g_m$

$$A_i = \frac{g_m R_G}{1 + g_m R_L}$$

3)

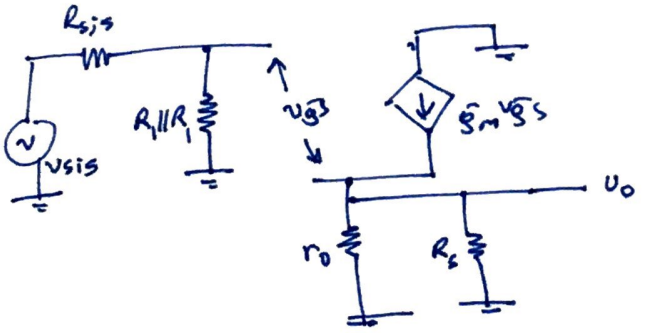


4)



(same analysis as (2))

5)



$$A_v = \frac{g_m R_s}{1 + g_m R_s}$$

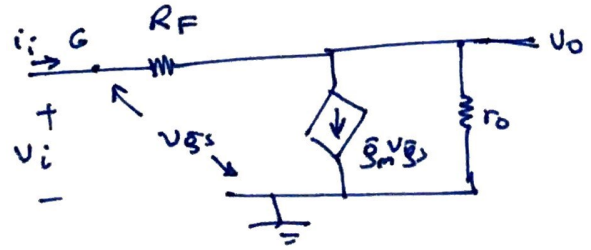
$R_{in} = R_1/2$

$R_o = \frac{1}{g_m} // R_s$

$$A_i = \frac{g_m R_s}{1 + g_m R_s} \cdot \frac{R_1/2}{R_L}$$

(whatever R_L may be)

6)



$v_i = v_{gs}$

$$\frac{v_o - v_i}{R_F} + g_m v_{gs} = 0 \quad \text{(ignore } r_o\text{)}$$

$$\Rightarrow \frac{v_o}{v_i} = 1 - g_m R_F$$

$i_i = g_m v_{gs} = g_m v_i$

$$\Rightarrow R_i = \frac{v_i}{i_i} = 1/g_m$$

$R_o = R_F$ (by inspection)

$$A_i = (1 - g_m R_F)\frac{1}{g_m R_L}$$

(whatever R_L may be)